

Bernoulli's differential equation

A diff. eq. of the form

$\frac{dy}{dx} + P(x)y = Q(x)y^n$ where n is a real number except 0 and 1 is called a Bernoulli's diff. eq.

Method Dividing the eq by y^n

$$\frac{1}{y^n} \frac{dy}{dx} + P(x) \frac{1}{y^{n-1}} = Q(x)$$

$$\text{i.e. } y^{-n} \frac{dy}{dx} + P(x) y^{1-n} = Q(x) \rightarrow (1)$$

$$\text{Put } z = y^{1-n} \quad \therefore \frac{dz}{dx} = (1-n) y^{-n} \frac{dy}{dx}$$

$$\text{i.e. } \frac{1}{1-n} \frac{dz}{dx} = y^{-n} \frac{dy}{dx}$$

$$\text{sub. in eq (1), } \frac{1}{1-n} \frac{dz}{dx} + P(x)z = Q(x)$$

$$\text{i.e. } \frac{dz}{dx} + (1-n)P(x)z = (1-n)Q(x)$$

This is a linear diff. eq. with z as dependent variable.

Solve the following diff. eqs.

$$(1) \frac{dy}{dx} + 2y = y^2$$

$$(2) x \frac{dy}{dx} + y = xy^3$$

$$(3) \frac{dy}{dx} - y \tan x = y^4 \sec x$$

$$(4) \frac{dy}{dx} + \tan x \cdot \tan y = \cos x \cdot \sec y$$

$$(5) \frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$$

Solutions (1) $\frac{dy}{dx} + 2y = y^2$

Dividing by y^2 , $\frac{1}{y^2} \frac{dy}{dx} + \frac{2}{y} = 1$

$$\text{i.e. } y^{-2} \frac{dy}{dx} + 2y^{-1} = 1 \rightarrow (1)$$

$$\text{Put } z = y^{-1}, \quad \frac{dz}{dx} = -1 y^{-2} \frac{dy}{dx}, \quad -\frac{dz}{dx} = y^{-2} \frac{dy}{dx}$$

Sub in (1) $-\frac{dz}{dx} + 2z = 1$

i.e. $\frac{dz}{dx} - 2z = -1$ This is a linear diff eq.

$P = -2$, $Q = -1$. I.F. = e^{-2x}

Solution $y(I.F) = \int (I.F) Q dx$

i.e. $z e^{-2x} = \int -e^{-2x} dx$
 $= -\frac{e^{-2x}}{-2} + \frac{C}{2}$

i.e. $2z e^{-2x} = e^{-2x} + C$
 $\frac{2}{y} e^{-2x} = e^{-2x} + C$

i.e. $\frac{2}{y} = 1 + C e^{2x}$ is the solution.

(2) $x \frac{dy}{dx} + y = x y^3$

Dividing by $x y^3$

$\frac{1}{y^3} \frac{dy}{dx} + \frac{1}{x y^2} = 1$ i.e. $y^{-3} \frac{dy}{dx} + \frac{1}{x} y^{-2} = 1 \rightarrow (1)$

Put $z = y^{-2} \Rightarrow \frac{dz}{dx} = -2 y^{-3} \frac{dy}{dx} \Rightarrow -\frac{1}{2} \frac{dz}{dx} = y^{-3} \frac{dy}{dx}$

$\therefore$ we get $-\frac{1}{2} \frac{dz}{dx} + \frac{1}{x} z = 1$

i.e. $\frac{dz}{dx} - \frac{2}{x} z = -2$, $P = -\frac{2}{x}$, $Q = -2$

I.F. = $e^{\int -\frac{2}{x} dx} = e^{-2 \log x} = e^{\log x^{-2}} = \frac{1}{x^2}$

Solution: $z(I.F) = \int (I.F) Q dx$

i.e. $z \cdot \frac{1}{x^2} = \int \frac{1}{x^2} (-2) dx$

i.e. $z \cdot \frac{1}{x^2} = 2 \int -\frac{1}{x^2} dx$
 $= 2 \left(\frac{1}{x} \right) + C$

i.e. $z = 2x + C x^2$

i.e. $\frac{1}{y^2} = 2x + C x^2$ is the solution.

$$(3) \frac{dy}{dx} - y \tan x = y^4 \sec x$$

Dividing by y^4

$$\frac{1}{y^4} \frac{dy}{dx} - \frac{1}{y^3} \tan x = \sec x$$

$$\text{or } y^{-4} \frac{dy}{dx} - y^{-3} \tan x = \sec x$$

$$\text{put } z = y^{-3}, \quad \frac{dz}{dx} = -3y^{-4} \frac{dy}{dx}, \quad -\frac{1}{3} \frac{dz}{dx} = y^{-4} \frac{dy}{dx}$$

$$\text{or } -\frac{1}{3} \frac{dz}{dx} - \tan x \cdot z = \sec x$$

$$\text{or } \frac{dz}{dx} + 3 \tan x \cdot z = -3 \sec x$$

$$P = 3 \tan x, \quad Q = -3 \sec x$$

$$I.F. = e^{\int 3 \tan x dx} = e^{3 \log \sec x} = e^{\log \sec^3 x} = \sec^3 x$$

Solution is given by

$$z \cdot \sec^3 x = -3 \int \sec^3 x \cdot \sec x dx$$

$$= -3 \int \sec^2 x \cdot \sec^2 x dx$$

$$= -3 \int (1 + \tan^2 x) \sec^2 x dx$$

$$= -3 \int (1 + t^2) dt$$

$$= -3 \left(t + \frac{t^3}{3} + \frac{C}{3} \right)$$

$$\left(\text{Put } t = \tan x \right. \\ \left. dt = \sec^2 x dx \right)$$

$$\text{or } \frac{1}{y^3} \sec^3 x = -3 \left(\tan x + \frac{\tan^3 x}{3} + \frac{C}{3} \right)$$

$$\text{or } \frac{\sec^3 x}{y^3} = -(\tan^3 x + 3 \tan x + C) \text{ is the soln.}$$

$$(4) \frac{dy}{dx} + \tan x \cdot \tan y = \cos x \cdot \sec y$$

Dividing by $\sec y$

$$\frac{1}{\sec y} \frac{dy}{dx} + \frac{\tan x \cdot \tan y}{\sec y} = \cos x$$

$$\text{or } \cos y \frac{dy}{dx} + \tan x \cdot \sin y = \cos x \rightarrow (1)$$

$$\text{put } z = \sin y, \quad \frac{dz}{dx} = \cos y \frac{dy}{dx}$$

$$i. \frac{dz}{dx} + \tan x \cdot z = \cos x$$

$$P = \tan x, \quad Q = \cos x$$

$$IF = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$$

$$\text{solution: } z (\sec x) = \int \sec x \cdot \cos x dx$$

$$i. z \sec x = \int 1 dx$$

$$\sin x \cdot \sec x = x + C$$

$$(5) \frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$$

Dividing by $\cos^2 y$

$$\frac{1}{\cos^2 y} \frac{dy}{dx} + x \frac{\sin 2y}{\cos^2 y} = x^3$$

$$i. \sec^2 y \frac{dy}{dx} + x (2 \tan y) = x^3$$

$$\left(\because \frac{\sin 2y}{\cos^2 y} = \frac{2 \sin y \cos y}{\cos^2 y} = 2 \tan y \right)$$

$$\text{Put } z = \tan y, \quad \frac{dz}{dx} = \sec^2 y \frac{dy}{dx}$$

$$\therefore \text{ we get } \frac{dz}{dx} + 2xz = x^3 \quad \text{which is a linear eq}$$

$$P = 2x, \quad Q = x^3$$

$$IF = e^{\int 2x dx} = e^{x^2}$$

$$\text{solution } z \cdot e^{x^2} = \int e^{x^2} \cdot x^3 dx$$

$$= \int e^{x^2} \cdot x^2 \cdot x dx$$

$$= \int e^u \cdot u \frac{du}{2}$$

$$= \frac{1}{2} \int u e^u du$$

$$= \frac{1}{2} (u e^u - \int e^u du)$$

$$i. \tan y e^{x^2} = \frac{1}{2} (u e^u - e^u) + \frac{C}{2}$$

$$i. 2 \tan y e^{x^2} = e^u (u - 1) + C$$

$$= e^{x^2} (x^2 - 1) + C$$

$$i. 2 \tan y = (x^2 - 1) + C e^{-x^2}$$

Solve the following

$$(1) \frac{dy}{dx} + y = xy^3 \quad (2) \frac{dy}{dx} - y \tan x = y^2 \sec x$$

$$(3) \frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x} \log x \quad (4) \frac{dy}{dx} + xy = x^3 y^3$$

Partial differentiation

If z is a function of x and y say $f(x, y)$ it can be differentiated partially w.r. to x or partially w.r. to y .

Partial derivative of z w.r. to x is defined as

$$\frac{\partial z}{\partial x} = \lim_{\Delta x \rightarrow 0} \left[\frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \right]$$

Here we treat y as a constant.

Partial derivative of z w.r. to y is defined as

$$\frac{\partial z}{\partial y} = \lim_{\Delta y \rightarrow 0} \left[\frac{f(x, y + \Delta y) - f(x, y)}{\Delta y} \right]$$

Here we treat x as a constant.

Problems

Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if

(i) $z = x^3 + 3x^2y + xy^3$

(ii) $z = \tan^{-1}\left(\frac{y}{x}\right)$

(iii) $z = 2x \cos y + 3x^2y$

(iv) $z = x^3 - x^2 \sin y - y$

(i) $z = x^3 + 3x^2y + xy^3$

$$\frac{\partial z}{\partial x} = 3x^2 + 6xy$$

$$\frac{\partial z}{\partial y} = 3x^2 + 3xy^2$$

(ii) $z = \tan^{-1}\left(\frac{y}{x}\right)$

$$\frac{\partial z}{\partial x} = \frac{1}{\left(1 + \frac{y^2}{x^2}\right)} \frac{\partial}{\partial x} \left(\frac{y}{x}\right) = \frac{x^2}{x^2 + y^2} \left(-\frac{y}{x^2}\right) = \frac{-y}{x^2 + y^2}$$

$$\frac{\partial z}{\partial y} = \frac{1}{\left(1 + \frac{y^2}{x^2}\right)} \frac{\partial}{\partial y} \left(\frac{y}{x}\right) = \frac{x^2}{x^2 + y^2} \left(\frac{1}{x}\right) = \frac{x}{x^2 + y^2}$$

$$(iii) z = 2x \cos y + 3x^2y$$

$$\frac{\partial z}{\partial x} = 2 \cos y + 6xy \quad \frac{\partial z}{\partial y} = -2x \sin y + 3x^2$$

Higher order partial derivatives

$$(i) \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) \quad (ii) \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right)$$

$$(iii) \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) \quad (iv) \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right)$$

If $z = f(x, y)$ is a continuous function
then $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$

∴ order of differentiation is immaterial

Find first and second order partial derivatives of the following functions

$$(i) x^2y$$

$$(ii) x^3 \sin y$$

$$(i) z = x^2y$$

$$\frac{\partial z}{\partial x} = 2xy, \quad \frac{\partial z}{\partial y} = x^2, \quad \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} (2xy) = 2y$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} (x^2) = 0, \quad \frac{\partial^2 z}{\partial y (\partial x)} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial y} (2xy) = 2x$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial x} (x^2) = 2x$$

Here we notice that $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$

$$(ii) z = x^3 \sin y$$

$$\frac{\partial z}{\partial x} = 3x^2 \sin y$$

$$\frac{\partial z}{\partial y} = x^3 \cos y$$

$$\frac{\partial^2 z}{\partial x^2} = 6x \sin y$$

$$\frac{\partial^2 z}{\partial y^2} = -x^3 \sin y$$

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = 3x^2 \cos y = \frac{\partial^2 z}{\partial x \partial y}$$

Differentials If $z = f(x, y)$ then dx, dy and dz are called differentials in x, y and z respectively

Differential in z is given by

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

Example

$$z = x^2 y$$

$$\begin{aligned} dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \\ &= 2xy dx + x^2 dy \end{aligned}$$

1) Find differential in z if $f(x, y) = \frac{x^3}{3} - xy^2$

$$\frac{\partial f}{\partial x} = x^2 - y^2, \quad \frac{\partial f}{\partial y} = 0 - 2xy = -2xy$$

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = (x^2 - y^2) dx - 2xy dy$$

Exact differential equation

A diff. eq of the form $M(x, y) dx + N(x, y) dy = 0$ is said to be an exact diff. eq if there exists a function $u(x, y)$ such that $\frac{\partial u}{\partial x} = M(x, y)$ and $\frac{\partial u}{\partial y} = N(x, y)$

In this case $M dx + N dy = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$

$$\text{i.e. } M dx + N dy = du$$

This is called an exact differential

It's solution is given by $u(x, y) = c$

Example consider the diff. eq

$$(1-x) dx - (1+y) dy = 0, \quad M = 1-x, \quad N = -1-y$$

$$\text{Let } u(x, y) = \left(x - \frac{x^2}{2}\right) - \left(y + \frac{y^2}{2}\right)$$

then $\frac{\partial u}{\partial x} = 1-x = M$ $\frac{\partial u}{\partial y} = -(1+y) = N$

Given diff. eq is exact.

Solution is given by

$$u(x,y) = C \Rightarrow x - \frac{x^2}{2} - (y + \frac{y^2}{2}) = C$$

Condition for exactness

consider the diff. eq

$$M(x,y) dx + N(x,y) dy = 0 \text{ which is exact}$$

$\therefore$ There is a function $u(x,y)$ such that

$$\frac{\partial u}{\partial x} = M \quad \text{and} \quad \frac{\partial u}{\partial y} = N$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial M}{\partial y} \quad \text{and} \quad \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial N}{\partial x}$$

$$\text{But } \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y} \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ A diff. eq of the form $M(x,y) dx + N(x,y) dy = 0$ is exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Solution of an exact diff. eq.

$$\text{Solve } (x^2 - y^2) dx - 2xy dy = 0$$

$$M = x^2 - y^2, \quad N = -2xy$$

$$\frac{\partial M}{\partial y} = -2y \quad \frac{\partial N}{\partial x} = -2y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \therefore \text{eq is exact}$$

There is a function $u(x,y)$ such that

$$\frac{\partial u}{\partial x} = M = x^2 - y^2 \quad \frac{\partial u}{\partial y} = N = -2xy \rightarrow (2)$$

$$u = \int M dx = \int (x^2 - y^2) dx = \frac{x^3}{3} - xy^2 + \phi(y) \rightarrow (3)$$

$$\text{u } u(x,y) = \frac{x^3}{3} - xy^2 + \phi(y)$$

diff w.r. to y

$$\frac{\partial u}{\partial y} = -2xy + \phi'(y) \quad \text{ie } N = -2xy + \phi'(y)$$

But $N = -2xy \quad \therefore \phi'(y) = 0 \quad \therefore \phi(y) = C$

sub in (1) $u(x, y) = \frac{x^3}{3} - xy^2 + C$

$\therefore$ Soln of the given diff. eq is

$$u(x, y) = C_1$$

$$\text{ie } \frac{x^3}{3} - xy^2 + C = C_1 \Rightarrow \underline{\underline{\frac{x^3}{3} - xy^2 = C_1}}$$

Short-cut method

solution is given by

$$\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$$

Example $(x^2 - y^2) dx - 2xy dy = 0$

$$M = x^2 - y^2 \quad N = -2xy$$

Soln: $\int (x^2 - y^2) dx + \int 0 dy = C$

$$\text{ie } \underline{\underline{\frac{x^3}{3} - xy^2 = C}}$$

Solve $(3x^2 + 4xy) dx + (2x^2 + 2y) dy = 0$

$$M = 3x^2 + 4xy, \quad N = 2x^2 + 2y$$

$$\frac{\partial M}{\partial y} = 4x, \quad \frac{\partial N}{\partial x} = 4x$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \therefore \text{Eq is exact}$$

Soln: $\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$

$$\text{ie } \int (3x^2 + 4xy) dx + \int 2y dy = C$$

$$\text{ie } \frac{3x^3}{3} + 4 \frac{x^2 y}{2} + y^2 = C$$

$$\text{ie } x^3 + 2x^2 y + y^2 = C$$